

Multi-Source Candidate Data Transformer — Technical Design

Design Philosophy

Rather than merging data as soon as it is parsed, I designed the pipeline around a "collect first, decide later" approach. Every source contributes evidence independently, and the final decision is made only after all available information has been collected and normalized. This keeps the transformation deterministic, explainable, and easy to extend without affecting existing components.

Data Processing Flow

Source Ingestion (Recruiter CSV, ATS JSON, Resume, Recruiter Notes) → Independent Parsing → Canonical Field Extraction → Normalization & Alias Resolution → Evidence Collection → Deterministic Merge Engine → Confidence & Provenance Generation → Profile Validation → Runtime Output Projection → Final JSON Output

Each source is processed independently. If a file is missing or malformed, it is logged and skipped while the remaining sources continue through the pipeline. This ensures the application always returns the best possible candidate profile instead of failing because of a single bad input.

Key Design Decisions

The Evidence Collection stage is the core of the design. Instead of merging records immediately, every extracted value is first grouped by its corresponding candidate field. This allows the merge engine to compare all available evidence before selecting the final value.

The canonical profile follows the assignment schema and stores normalized identity, contact details, skills, work experience, education, confidence scores, and provenance. During normalization, emails are lowercased, phone numbers follow the E.164 format, dates use the YYYY-MM standard, and skills are converted into their canonical representation while preserving unknown values.

Whenever different sources provide conflicting information, the merge engine follows a deterministic policy. It first prefers values confirmed by multiple independent sources. If there is no clear consensus, source reliability is used as the deciding factor (Recruiter CSV → ATS JSON → Resume → GitHub → LinkedIn → Recruiter Notes). Any remaining ties are resolved deterministically so that identical inputs always generate identical outputs.

Instead of assigning equal importance to every source, confidence scores are calculated using both source reliability and cross-source corroboration. Provenance is maintained for every merged field by recording where the selected value originated and how it was extracted, making the final profile transparent and easy to audit.

The system always creates a single canonical profile internally. A separate Projection Engine then converts this profile into either the default schema or a custom output defined through a runtime JSON configuration. This allows fields to be renamed, nested, omitted, or validated without modifying the transformation logic.

To keep the solution modular, I separated parsing, merging, validation, and projection into independent components using the Strategy, Factory, and Pipeline design patterns. As a result, supporting a new data source only requires implementing another parser, while the merge, confidence, validation, and projection layers remain unchanged.